

# Human-Visual-System-Inspired Underwater Image Quality Measures

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**Abstract**—Underwater images suffer from blurring effects, low contrast, and grayed out colors due to the absorption and scattering effects under the water. Many image enhancement algorithms for improving the visual quality of underwater images have been developed. Unfortunately, no well-accepted objective measure exists that can evaluate the quality of underwater images similar to human perception. Predominant underwater image processing algorithms use either a subjective evaluation, which is time consuming and biased, or a generic image quality measure, which fails to consider the properties of underwater images. To address this problem, a new nonreference underwater image quality measure (UIQM) is presented in this paper. The UIQM comprises three underwater image attribute measures: the underwater image colorfulness measure (UICM), the underwater image sharpness measure (UISM), and the underwater image contrast measure (UIConM). Each attribute is selected for evaluating one aspect of the underwater image degradation, and each presented attribute measure is inspired by the properties of human visual systems (HVSs). The experimental results demonstrate that the measures effectively evaluate the underwater image quality in accordance with the human perceptions. These measures are also used on the AirAsia 8501 wreckage images to show their importance in practical applications.

**Index Terms**—AirAsia 8501 wreckage search, colorfulness measure, contrast measure, port security, sharpness measure, underwater image quality evaluation.

## I. INTRODUCTION

THE quality of underwater images plays a pivotal role in ocean engineering and scientific research, such as mon-

uating the image processing algorithms' performances. However, subjective evaluation is biased, and it is expensive with respect to time and resources. More importantly, subjective evaluation cannot be automated. Therefore, it is important to have a reliable objective evaluation measure. Generally, objective evaluation methods can be classified into two categories: non-reference measures and full-reference measures, depending on whether the images with the "true color" and "ideal contrast" are available [8]. In underwater image processing scenarios, such ideal images are usually unavailable. Therefore, the non-reference image quality measures are desired. This challenge is further exacerbated by the need to develop an underwater image quality measure that captures the objectivity and perception of the human visual system (HVS). Some generic image quality measures, such as the receiver operating characteristic

(ROC) curve analysis [6], the histogram analysis [9], [10], the Michelson contrast measure [11], Hasler's colorfulness measure [12], and color image quality measure (CQI) [13] have been developed. However, these objective measures are not designed specifically for underwater images. They fail to consider the strong absorption and scattering effects of the water and they are not able to provide a comprehensive evaluation for underwater images.

Recently, we have shown that the overall quality of an image can be effectively obtained by combinations of image attribute measures. It is natural to investigate how to tailor the image attribute measures for use in evaluating the overall underwater image quality. For underwater images, the different absorption rate for light with different wavelengths causes color casting,

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dient structural similarity measure (GSSIM) [19] that attempt to quantify the difference perceptually. The UIQM does not require a reference image and it utilizes the HVS model which correlates well with the perceived underwater image quality. The remainder of this paper is organized as follows. Section II briefly presents the characteristics of HVS. Quality measures used in existing underwater image processing algorithms are also reviewed in this section. Section III presents detailed designs of the new colorfulness, sharpness, contrast measures, as well as the overall quality measure for underwater images. Section IV shows the experimental results of the presented measures for evaluating underwater image attributes and qualities. The underwater wreckage images obtained during the AirAsia 8501 blackbox retrieval are used to show the practical applications of the measures in ocean rescue. Finally, the conclusions are discussed in Section V.

## II. BACKGROUND

The goal of the new objective measures is to achieve a high correlation with human perception of image quality. To achieve this, HVS-inspired underwater image attributes measures are presented. In this section, some essential HVS phenomena that are pertinent in developing the underwater image quality measures are reviewed. Other current existing objective evaluation methods are also reviewed in this section.

### A. Human Visual System Characteristics

We choose the underwater image colorfulness, sharpness, and contrast as the basic attributes for measuring. These attributes were chosen not only for their susceptibility to change due to the physical property of the water media, such as where absorption causes color desaturation and color casting, scattering causes blurring and contrast degradations, but also because the HVS is sensitive to the changes in color, edge structures, and relative contrast [14]. That is to say, changes in these image attributes result in completely different visual appearances according to

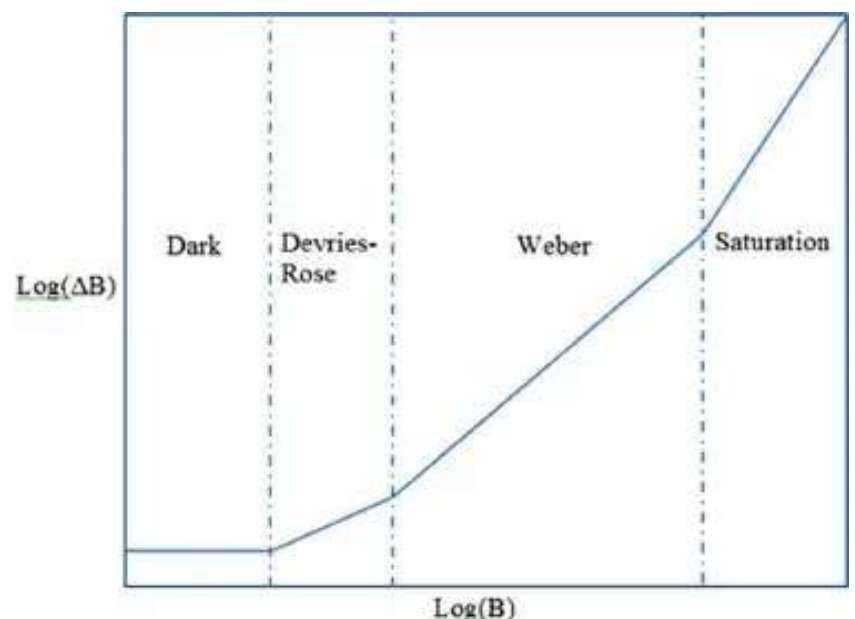


Fig. 1. Buchsbaum curve: perceived contrast ( $\Delta B$ ) is divided into four regions: dark region, Devries–Rose region, Weber region, and saturation region according the background intensity ( $B$ ).

background illumination level dependent. Accordingly, our underwater image contrast measure is formulated as a function of the relative contrast change. Edge sharpness can be seen as the contrast over the edge pixels, so a measure of the relative change is also adopted in the sharpness evaluation.

The parameterized logarithmic image processing (PLIP) provides the nonlinear representation and processing operations which are consistent with many HVS properties [26], such as brightness range inversion, Weber's Law, saturation characteristics, and context-dependent psychophysical notation [27]. The PLIP operations are also accounted for in the formulation of the underwater contrast measure.

### B. Objective Measures Used in Existing Underwater Image Processing Algorithms

Currently, objective evaluation for underwater images is still a challenge. The majority of underwater image processing al-

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overcome this limitation, some methods assume that the distribution of the gradient magnitude histogram is exponential except for a small peak at low gradient corresponding to homogeneous background [30]. Consequently, closeness of the resultant histogram to the exponential distribution is used to indicate the contrast improvement [9], [10]. The major limitation of these measures is that some important characteristics, such as colorfulness and color cast, are only accounted for indirectly.

### III. DEVELOPMENT OF THE NEW UNDERWATER IMAGE QUALITY MEASURE UIQM

The new underwater image quality measure UIQM comprises three attribute measures, namely, the underwater image colorfulness measure (UICM), the underwater image sharpness measure (UISM), and the underwater image contrast measure (UIConM). Details of the new measures are presented in this section.

#### A. Underwater Image Colorfulness Measure (UICM)

Many underwater images suffer from a severe color-casting problem. As the depth of the water increases, colors attenuate one by one depending on their wavelength [2]. The color red disappears first due to it possessing the shortest wavelength. As a result, underwater images usually demonstrate a bluish or greenish appearance. Furthermore, limited lighting conditions also causes severe color desaturation in underwater images. A good underwater image enhancement algorithm should produce good color rendition.

Research on natural scene colorfulness shows that colorfulness can be represented effectively with functions of image statistical values [12], [31]. Practically, users have the flexibility to choose which color spaces and which order statistical values to use, as well as the design of the fusion functions and the weighting coefficients. As mentioned before, the HVS captures colors in the opponent color plane. Therefore, the two opponent color components related with chrominance RG and YB



Fig. 2. (a) Underwater image with air bubbles [33]. (b) Underwater image with natural scenes [29]. In (a), the 20% brightest pixels are excluded from the calculation of the mean values because the bubbles are not related with the real scene. In (b), all pixels are considered in the calculation of the asymmetric alpha-trimmed means because this is a noise-free natural underwater scene image.



Fig. 3. (a) Original underwater image. (b) Color-corrected underwater image. Images courtesy of [34]. It is seen that the original image shows strong green tone, while the color-corrected image recovers more vivid color. The UICM assigns a greater measure value for the color-corrected image.

The asymmetric alpha-trimmed mean is applicable for different types of underwater images when parameters  $\alpha_L$  and  $\alpha_R$  change. Generally, for underwater images with lots of bubbles or light rays, such as the image in Fig. 2(a), a greater  $\alpha_R$  should be used to remove the effect of the bright regions, which are not related with the natural underwater scenes, as well as some high-frequency noisy pixels, such as the salt and pepper

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TABLE I  
STATISTICAL VALUES FOR IMAGES IN FIG. 3

|            | $\mu_{RG}$ | $\mu_{YB}$ | $\sigma_{RG}$ | $\sigma_{YB}$ | UICM    |
|------------|------------|------------|---------------|---------------|---------|
| Fig. 3 (a) | -10.8300   | 61.8086    | 13.1411       | 11.0085       | -0.5889 |
| Fig. 3 (b) | 0.7896     | 0.8901     | 23.9203       | 19.9269       | 4.9058  |

demonstrates the pixel activity within each color component. Intuitively, a greater variance corresponds to a higher dynamic range. Therefore, the differences in color are more distinguishable. Table I summarizes the variances of the  $RG$  and  $YB$  color components for the original image in Fig. 3(a) and the color-enhanced image in Fig. 3(b). It is observed that the enhanced image in Fig. 3(b) has more vivid color, which correspond to greater variance values.

The overall colorfulness metric used for measuring underwater image colorfulness is demonstrated in

$$\text{UICM} = -0.0268\sqrt{\mu_{\alpha, RG}^2 + \mu_{\alpha, YB}^2} + 0.1586\sqrt{\sigma_{\alpha, RG}^2 + \sigma_{\alpha, YB}^2} \quad (5)$$

The linear combination coefficients are obtained by linear regression. It is seen that the coefficient for the mean term is negative and the coefficient for the variance is positive, which confirm the analysis of the mathematical meaning of the two terms.

Table I shows the sensitivity of the UICM for an underwater image in Fig. 3. Fig. 3(a) suffers from severe green color casting problem. The fusion-based enhancement algorithm in [34] effectively removes the background casting as shown in Fig. 3(b). The UICM assigns a greater colorfulness measure value for the enhanced image. The results show that with image statistical values, the UICM successfully measures the colorfulness attribute for underwater images.

### B. Underwater Image Sharpness Measure (UISM)

Sharpness is the attribute related to the preservation of fine details and edges. For images captured under the water, severe

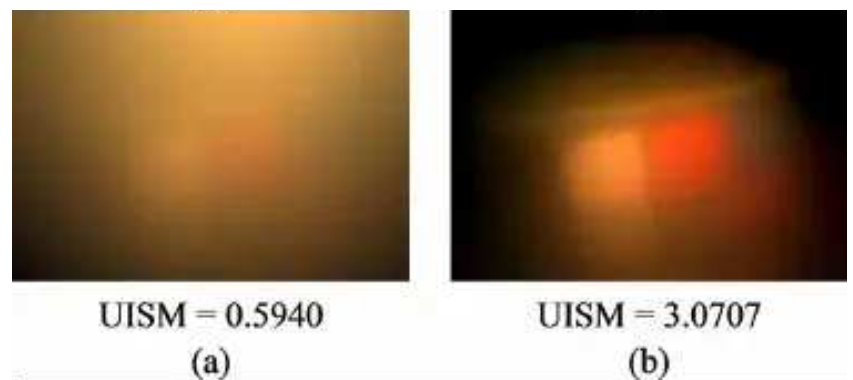


Fig. 4. (a) Original underwater image. (b) Descattered underwater image. Images courtesy of [36]. It is seen that the original image suffers from severe blurring effect, while the descattered image has sharper edges. The UISM assigns a greater measure value for the descattered image.



Fig. 5. (a) Original underwater image. (b) Contrast-enhanced underwater image. Images courtesy of [34]. It is seen that the original image in (a) has low contrast, while the contrast is greatly enhanced in (b). The UIConM assigns a greater measure value for the contrast-enhanced image.

component are combined linearly with coefficients  $\lambda_c$ , where  $\lambda_R = 0.299$ ,  $\lambda_G = 0.587$ , and  $\lambda_B = 0.114$  are used according to the relative visual responses of the red, green, and blue channels [30].

The descattering algorithm in [36] aims to remove the scattering effect of water. It is seen from Fig. 4 that the descattered underwater image in Fig. 4(b) achieves better sharpness than the original underwater image in Fig. 4(a). Accordingly, the descattered image in Fig. 4(b) has a greater UISM value.

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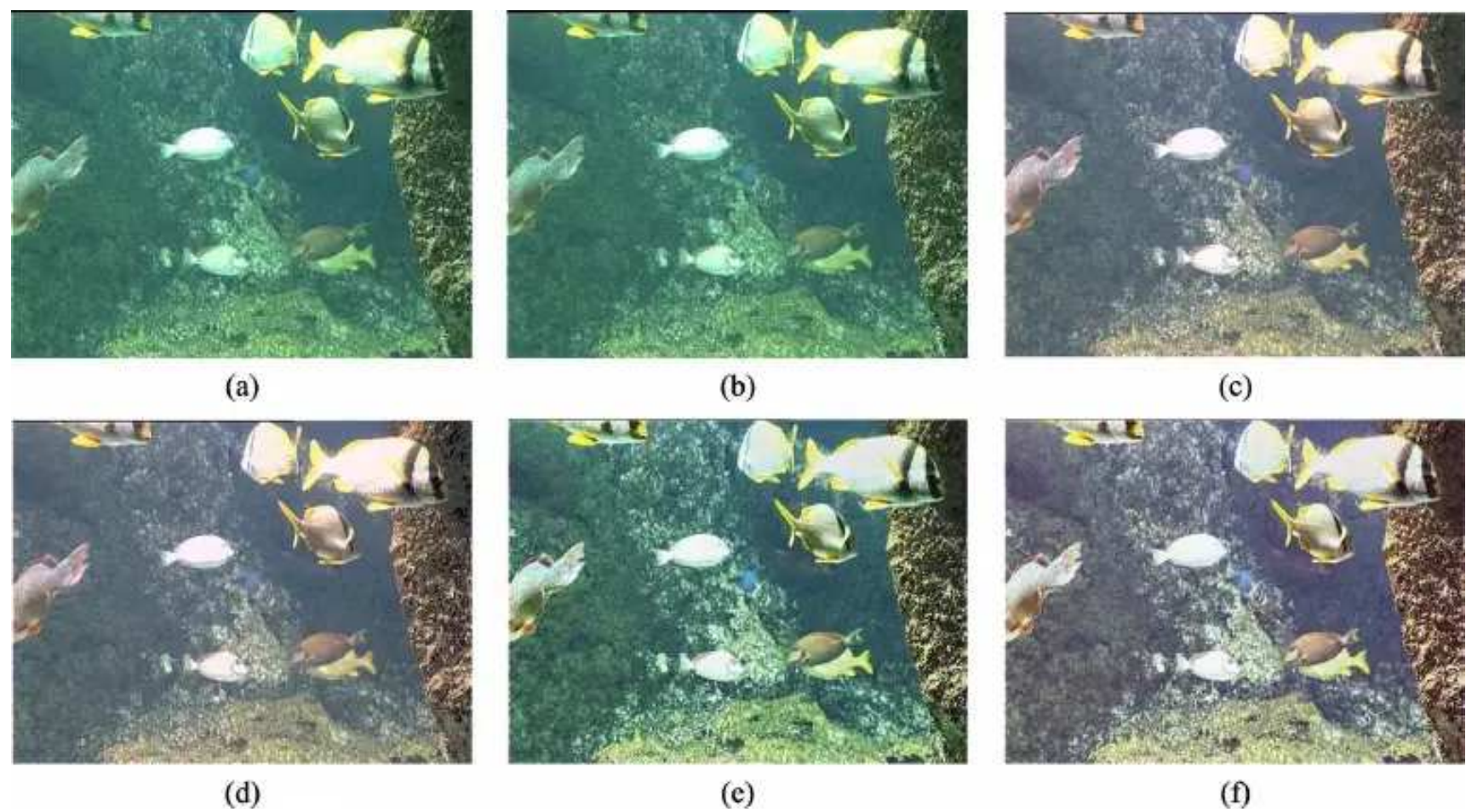


Fig. 6. (a) Original image. Color-enhanced images using (b) WP; (c) GW; (d) GWWP; and (e) ACE with original gray level preserved. (f) ACE without original gray level preserved. Images courtesy of [39].

TABLE II  
COLORFULNESS MEASURE VALUES FOR IMAGES IN FIG. 6

|            | UICM          | Hasler [12]    | CIQI [13]     |
|------------|---------------|----------------|---------------|
| Fig. 6 (a) | 0.9657        | 13.5015        | 0.4709        |
| Fig. 6 (b) | 0.9709        | 13.5205        | 0.4711        |
| Fig. 6 (c) | 2.4832        | 7.9950         | 0.3400        |
| Fig. 6 (d) | 2.4604        | 7.9666         | 0.3337        |
| Fig. 6 (e) | 2.1758        | <b>14.0897</b> | <b>0.4846</b> |
| Fig. 6 (f) | <b>2.7607</b> | 10.2331        | 0.3484        |

D. Underwater Image Quality Measure (UIQM)

It has been demonstrated that underwater images can be modeled as linear superposition of absorbed and scattered components [38]. Besides, it is known that the absorption and scattering effects cause color, sharpness, and contrast degradation. Therefore, it is reasonable to use the linear superposition model for generating the overall underwater image quality measure as well. The overall underwater image quality measure is then given by

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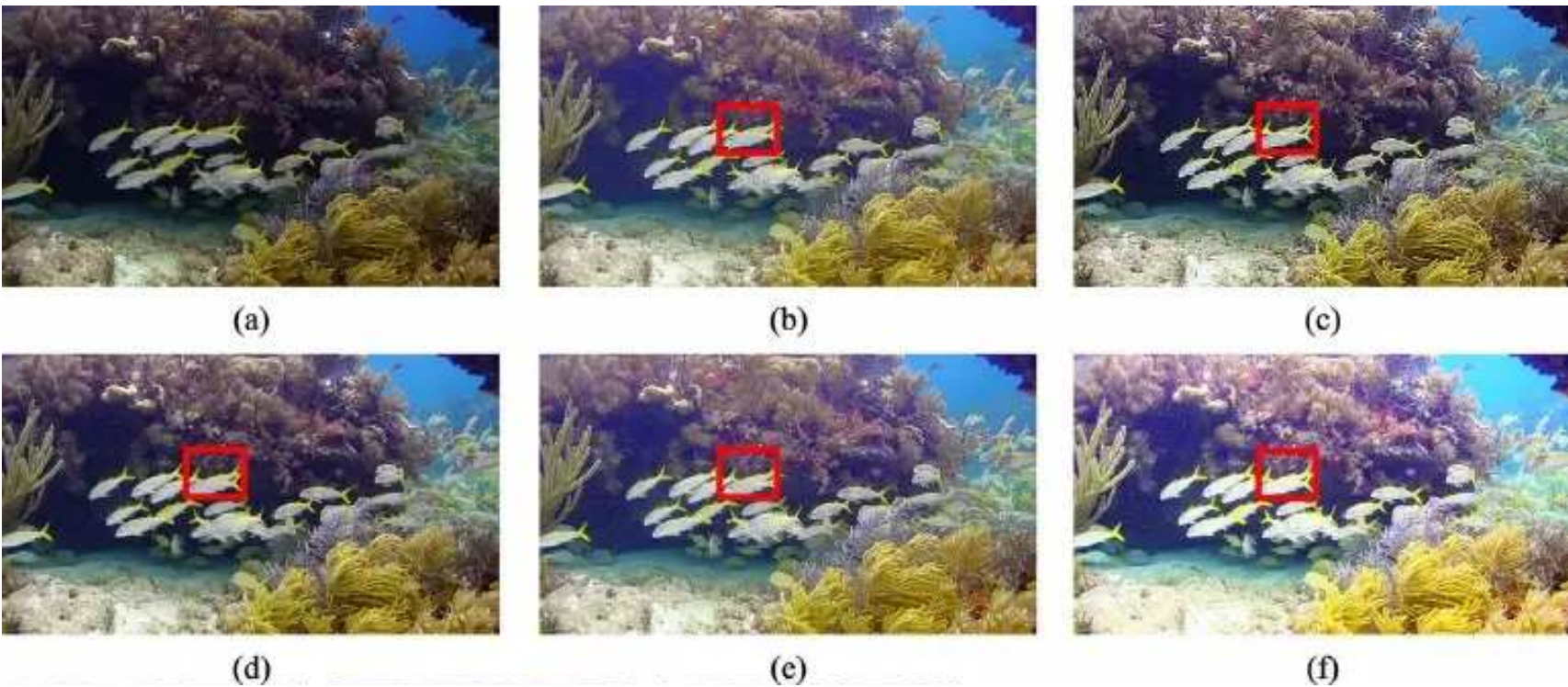


Fig. 7. (a) Original image. Enhancement results from (b) DRC\_CES; (c) MCE; (d) MCE\_DRC; (e) SF\_CES; and (f) TW\_CES. The original image is obtained from the SUN database [44].

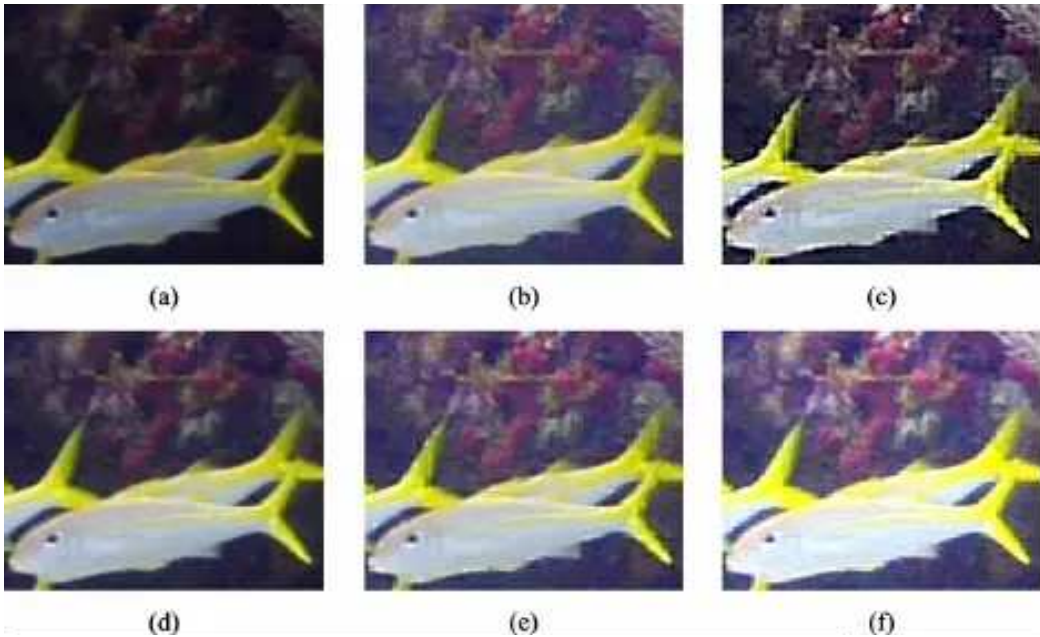


Fig. 8. Zoomed in view of the region of interest for the images from Fig. 7. (a) Original image. Enhancement results from (b) DRC\_CES; (c) MCE; (d) MCE\_DRC; (e) SF\_CES; and (f) TW\_CES.

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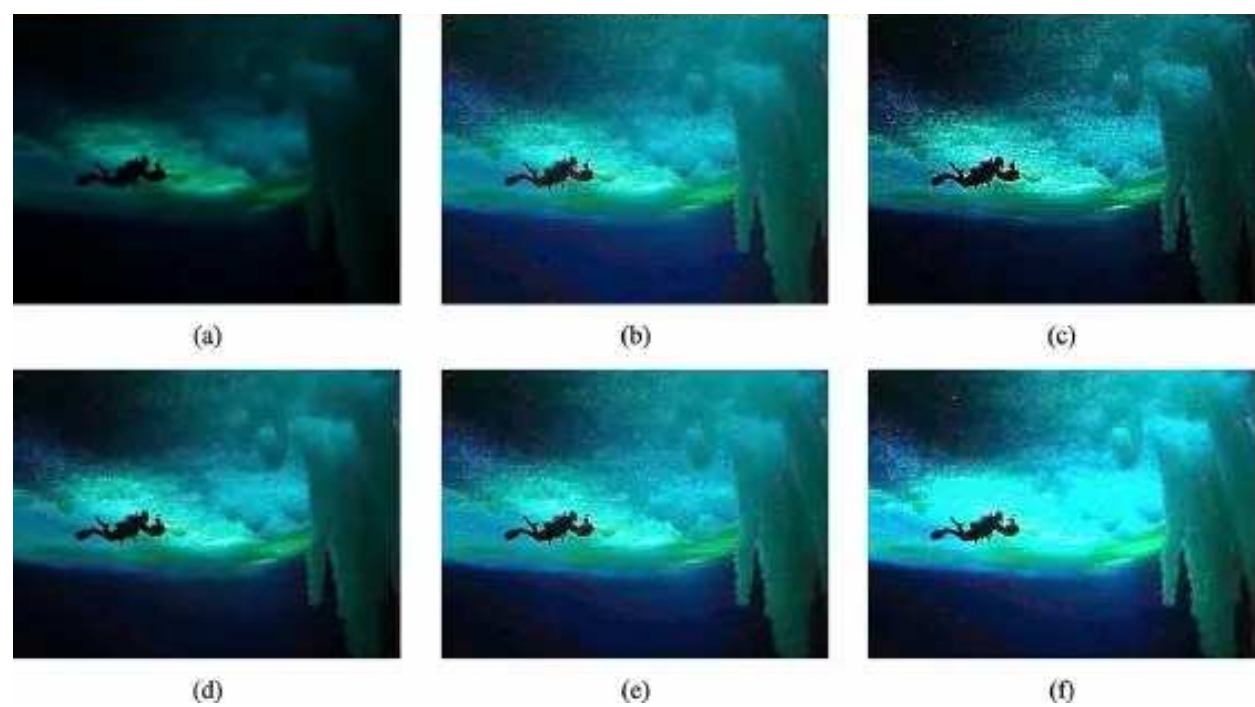


Fig. 9. (a) Original image. Enhancement results from (b) DRC\_CES; (c) DRC\_CES\_BLK; (d) MCE\_DRC; (e) SF\_CES; and (f) TW\_CES. The original image is obtained from the SUN database [44].



Fig. 10. The fourteen images used to test the performances of the quality measures.

UICM, Hasler’s color measure [12], and the CIQI colorfulness measure [13] are used for comparison purposes. Contrarily, these two methods select the “ACE method with the original measure [13] values are listed in Table II. Comparing these

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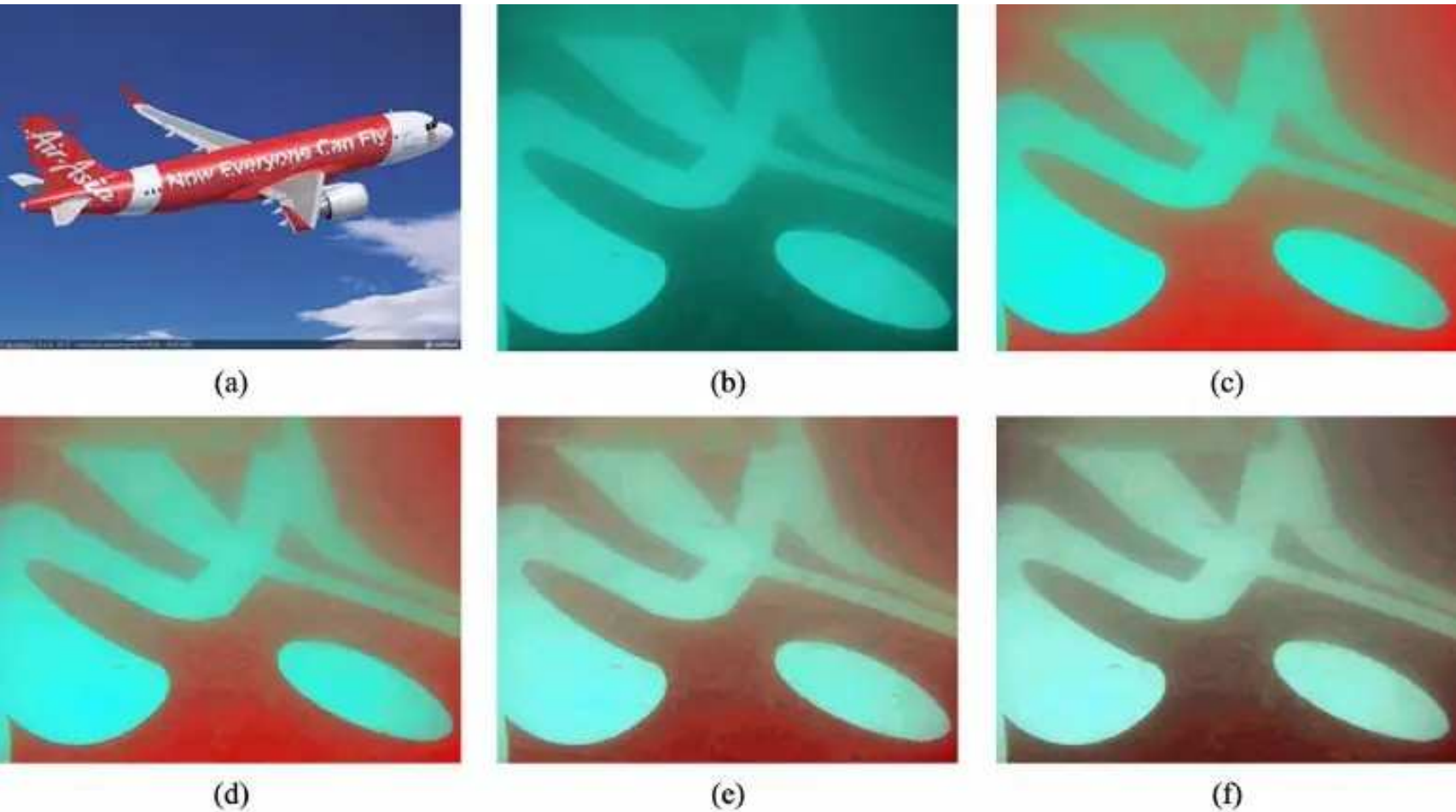


Fig. 11. (a) AirAsia plane. (b) Image from the wreckage of the AirAsia 8501 plane’s tail found underwater. (c)–(f) Color-enhancement results using the 3DAWQF with parameter  $M$  equal to 12, 20, 28, and 36, respectively.

TABLE VI

ATTRIBUTE AND QUALITY MEASURES FOR THE IMAGES IN FIG. 13. THE OPTIMAL RESULTS CORRESPONDING TO EACH MEASURE ARE HIGHLIGHTED IN BOLD, AND THE WORST RESULTS ARE HIGHLIGHTED IN ITALIC AND UNDERScore

|             | UICM   | UISM          | UIConM        | UIQM          |
|-------------|--------|---------------|---------------|---------------|
| Fig. 10 (c) | 9.4518 | 0.6242        | 0.4259        | 0.5062        |
| Fig. 10 (d) | 6.7425 | <b>1.0281</b> | <b>0.5666</b> | <b>0.6462</b> |
| Fig. 10 (e) | 6.7983 | 0.6217        | 0.4243        | 0.4854        |
| Fig. 10 (f) | 6.6958 | 0.7221        | 0.4722        | 0.5361        |
| Fig. 10 (g) | 4.9047 | <u>0.5995</u> | <u>0.3598</u> | <u>0.4108</u> |



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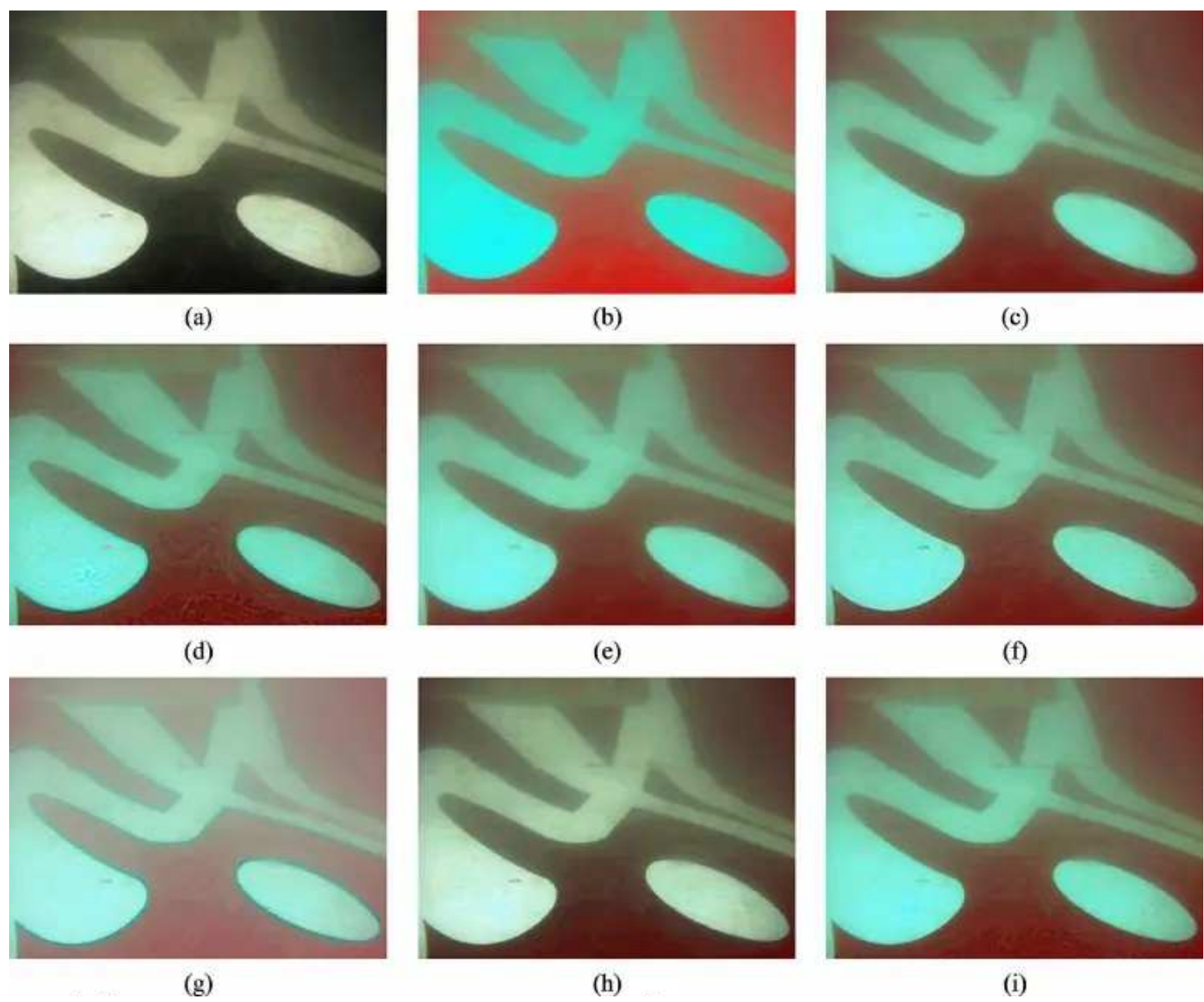


Fig. 13. Using presented underwater image attribute measures and underwater quality measures to select the optimal fusion method used in the 3DAWQF image-enhancement algorithm. (a) Intensity-corrected and contrast-enhanced image. (b) Color-enhanced image. (c) Filter-subtract-decimate pyramid fusion. (d) Contrast

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erated underwater vehicle during the retrieval of the blackbox from the AirAsia 8501 disaster. It is seen that the real color of the plane's tail [Fig. 11(a)] is not recognizable from the captured underwater image because the image shows a strong blue casting. In the 3DAWQF algorithm, a highpass filter is applied across the RGB color planes to recover colors. In the highpass filter used in 3DAWQF, a parameter  $M$  is used to control the influence of neighboring pixels to the center pixel of the color cube [40]. Fig. 11(c)–(f) shows color enhancement results with variant parameter  $M$ . It is seen that more color information is restored with a smaller value of  $M$ . A greater value of  $M$  also introduces some artifacts. The values of the UICM for parameter  $M$  ranging from 12 to 40 (Fig. 12) confirm the visual observations that a smaller  $M$  corresponds to a better color quality. Therefore, a parameter  $M$  equal to 12 is used in the 3DAWQF design.

Similarly, the UIConM can be used to assist the selection of optimal operating parameters used in the 3DAWQF contrast enhancement. After the optimal contrast-enhanced image [Fig. 13(a)] and the color-enhanced image [Fig. 13(b)] are obtained, these two images are fused together to generate the final output. Multiple fusion methods can be used [49], and fusion results from the filter-subtract-decimate pyramid, the contrast pyramid,

the gradient pyramid, the Laplacian pyramid, the morphological difference pyramid, the PLIP addition [26], and the ratio pyramid are shown in Fig. 13(e)–(i). The measure values for these fused images are tabulated in Table IV. From these measures, it is seen that the ratio pyramid fusion [Fig. 13(i)] has the most vivid color, while the PLIP fusion [Fig. 13(h)] has the least vivid color. The contrast pyramid fusion [Fig. 13(d)] has the best contrast, while the morphological difference pyramid fusion [Fig. 13(g)] has the worst contrast. The overall quality from the contrast pyramid fusion [Fig. 13(d)] is the best. These measures conform to the human perceptions. It is worth noting that for flight wreckage searching applications, similar aircraft models have similar painting colors. Therefore, the real painting colors are used in this section to validate the performance of the

underwater image quality measure UIQM can be effective tools to assist in selecting optimal operating parameters used in underwater enhancement algorithms as well as to rank algorithms' performances.

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